Mathematical overview: facts, assumptions and limits

For two **distinguished, fixed rows** of an infinite slotted border, a matching slide must be a period of each row. If their primitive periods are positive integers a and b slots, the shortest common matching slide is lcm(a,b). Distinguishing rows prevents exchanging the layers. The specified asymmetric F-like footprint has no matching nonidentity rotation; all footprints in Problems 1–2 have the same form and orientation, so matching slides are determined by their color words. Only the endlessly repeated pattern counts, not strip ends, construction brackets or R/B recording letters.

For recoloring a cyclic word so shift s becomes a symmetry, positions lie in cycles of i -> i+s modulo n. Each cycle must have a single color. Its minimum edit count is its size minus its largest original color class. Adding these independent lower bounds gives an attainable global minimum. A requested shift may work even when a smaller shift also works; Problem 2 does not demand a primitive period equal to s.

Two distinct forms can alternate around a necklace exactly when its number of cards is even. An odd necklace needs at least one equal-neighbor join, and one suffices. This is a cyclic adjacency statement, not a classification of planar necklace isometries or of seven frieze groups. Cards are symbolic holders of two forms, and they may be turned to suit handling; their adjacency is the object being studied.

Children discover common periods, optimal edits and a closure obstruction. Grades 2–3 have concrete slide/edit entries; Grades 4–5 can explain orbit lower bounds and arbitrary periods. K–1 may enter Problem 3 with adult reading and card placement. Experiments suggest rules; repeated-word/orbit arguments and alternating closure explanations establish them.

Preparation, readiness and launch

This companion adds three investigations; it is not a prescribed extra hour. A first visit can use P1 for 15–25 minutes and start P2; a later visit can devote 15–30 minutes to each edit family or to necklace constructions. These are unpiloted estimates. Keep the fixed K–1 / 2–3 / 4–5 tables. Use P3 at K–1; P1–2 at the third-grade table; offer all three and the general explanation at 4–5. Reading can be adult-supported. P1 needs repeating a 2–4-card word, counting slots to 12, and comparing two fixed rows. P2 needs keeping one original word intact, substituting colors, and counting changed positions; fluent division or lcm notation is unnecessary. P3 needs two-way alternation and counting to eight.

Per pair: 48 same-orientation footprint cards with reusable red/blue labels, or 48 red **and** 48 blue footprint cards if colors are fixed; 8 A and 8 mirror-A outline cards; two long reusable strips with 24 slots at 30 mm spacing (join paper lengths), two tracing strips, a ruler, pencil, and 8 removable join markers. Either color supply permits any assignment to the two 24-slot strips; the supplied RBB/RBBB start uses 14 red and 34 blue, and the RB/RBBBBBBB witness uses 15 red and 33 blue. A tracing copy records the pattern rather than using a second card inventory. Cards can be 25 mm long with their form and color clearly distinguished. Reuse Week 35 cards or trace the provided outline. Prepare pieces and durable strips before class; no child cutting is needed. For the eleven-child roster budget six pair-sized kits (two for each younger table, two at 4–5 so one can be demonstration/referee): 288 recolorable cards, or 288 red + 288 blue fixed-color cards; 48 A + 48 mirror-A cards, 12 slotted strips, 12 tracing strips, 6 rulers and 48 join markers. Six reusable-card kits are the lower-preparation option. The K–1 kits may omit the strips and color labels when using P3 alone.

Build P3 necklaces directly on the table with neighboring card centers at least 30 mm apart; leave the cards free to move rather than trying to fit them to the printed circle sites. Those 11.3 mm circles are symbolic records: children draw a small A or mirror-A mark, or the corresponding letter, after making a necklace. The large physical necklace and small drawing describe the same cyclic order. A partner can place the removable join markers between actual cards. The diagrams are not placement boards or cutting templates.

Allow two minutes of handling. Show a non-task red/blue repeat RB on one strip. Trace it, slide the copy two slots, and keep the original fixed. Point out that the short printed word is repeated in both directions. Before P2, use the student's non-task RBB → RRB → RRBRRB visual: one position changes in the block and the same edit recurs in every copy. Keep an original block intact beside a repaired

block, and count only the changed positions within that block. Before P3, join the ends of a four-card ABAB example on the table and identify the last–first neighbor pair; record the same cyclic order with four small drawn symbols. This small closure convention does not give away the odd-case obstruction.

Physical fit, strip joins, mirror-card handling and child transfer have **not** been pretested. Diagrams and code are digital checks. The material kit does not require exact pattern-block fit. One anchored adult at each table can read, preserve state and help with tracing; adult help should not choose a repair or route on the children's behalf.

Solutions, hints and continuations

P1 / student p. 1, two-row matching periods. Upper row RBB has primitive period 3 and lower RBBB primitive period 4; their first shared slide is 12 slots. R marks occur every third/fourth position, ruling out shorter primitive slides. A 6-slot pair can use RB and RBB; an 8-slot pair can use RB and RBBBBBBB. Alternatives can have both primitive periods equal to 6 or 8. Verify the generating rule, not only an interior window. Optional hint: keep one row's match while checking whether the other also matches. Extension: design two rows with matching individual slides but no shared slide smaller than their product; compare pairs whose lengths share a divisor. Ask children to justify a chosen shortest distance rather than merely name an lcm.

P2 / student p. 2, minimum recoloring. Treat each displayed word cyclically and make the same edits in every repeat. The cost counts positions changed within one displayed block; it does not count the infinitely many individual footprints in the endless border. RBBBRB under shift 3 pairs positions (1,4),(2,5),(3,6). The first two pairs disagree, giving minimum 2; all optimum words are RRBRRB, RBBRBB, BRBBRB, BBBBBB. RBBBBRRB under shift 2 has odd/even position cycles. The odd cycle has two R and two B, costing 2; the even cycle has three B and one R, costing 1. Minimum 3, with RBRBRBRB or BBBBBBBB. RRBBRB under shift 2 gives two three-position cycles. Each costs 1; the unique optimum is RBRBRB, minimum 2. Optimal repairs can create extra symmetries. Hint: follow where one position goes under repeated requested slides; use a string ring or partner pointing before introducing the word “cycle.” Extension: permit three colors, or prescribe two slides and compare their merged position groups. A new number word alone is a variant of this investigation, not a fourth counted investigation.

P3 / student p. 3, closing an alternating rule. Six and eight cards close with zero equal-neighbor joins; five and seven cannot. Starting at A and alternating, an odd final card is A too, so the seam joins equal forms. One equal join is attained by alternating every other join. For any binary ring, the number of changes of form is even: returning to the start changes its form an even number of times. Thus equal joins have the same parity as the card count. K–1 can give the last-card explanation with objects; the parity formulation is optional upper-band continuation. Further question: allow three forms and still require unequal neighbors; all rings of size at least three become possible. Do not count this small changed rule as a separate delivered investigation.

Distinction, provenance and checks

Base Week 35 K–1/2–3/4–5 packets explore individual repeating borders, glides, broken matching motions and composition/classification (see base Problems 1–6). New P1 keeps two layers distinguished and designs common periods; P2 optimizes fixed-position recolorings rather than moving footprints to break a motion; P3 changes the global object to a closed neighbor cycle. None reproduces Week 1 pattern-block games, cube-stack tilings or existing back-pocket extensions.

These are original finite designs, informed by the existing Week 35 mathematical overview and the Week 1 encore's familiar-material/new-question structure. No downloaded problem is copied. The outline cites Math Circle by the Bay, printed pp. viii–x, for flexible depth; that is a planning reference, not a source of these particular problems. student/verify.py checks primitive periods, exhaustive binary minimum repairs (all possibilities), and all binary necklaces of lengths 5–8. It establishes those finite examples; the overview explains generalization. Sources are portable; student/build.sh OUTPUT_DIR rebuilds bonus.pdf without repository imports. All pages are unpiloted; recording use of W35-B1, B2 or B3 with actual questions attempted remains future organizer work.